

Experiment - 2

TestDisk: Open-Source Data Recovery Tool

recover a missing partition and repair a corrupted one using test disk

Installation

Linux : `sudo apt-get install testdisk`

macOS : `brew install testdisk`

Windows: Download the executable from the official website.

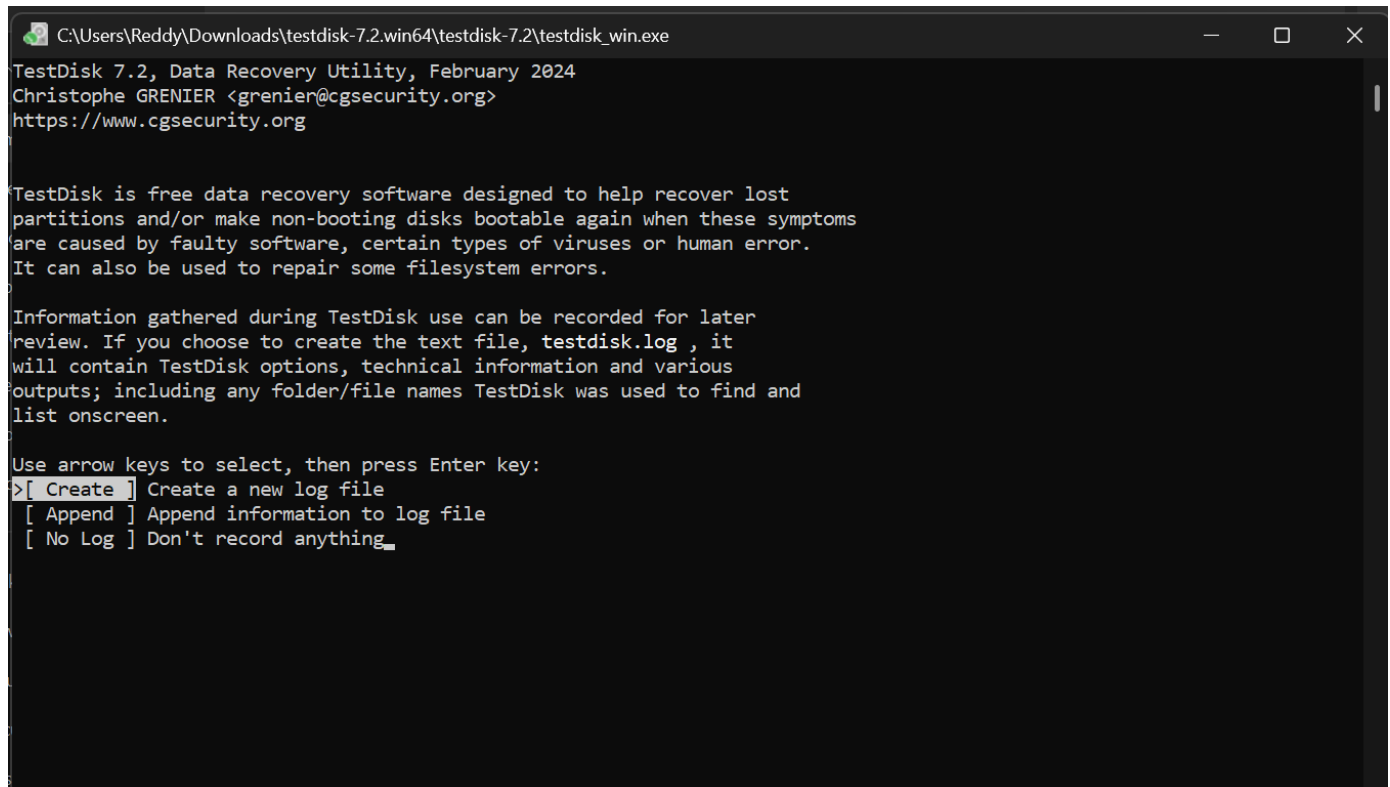
Link : https://www.cgsecurity.org/wiki/TestDisk_Download

NOTE:- if you using in windows please extract the .zip file first

Procedure

1. Create a Log File

- Launch TestDisk from your terminal or command prompt using `sudo testdisk` (or `testdisk_win.exe` on Windows).
- Select the [Create] option to generate a log file of the recovery session. This is helpful for future reference or troubleshooting.



```
C:\Users\Reddy\Downloads\testdisk-7.2.win64\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
>[ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything_
```

2. Select the Drive to Analyze

- TestDisk will display a list of all detected storage devices.
- Use the Up/Down arrow keys to highlight the drive that contains your lost data.
- Select [Proceed] to move to the next step.
-

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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - Micron MTFDKCD512TGE-1BK1AABLA

>[Proceed] [Quit]

Note: Serial number 00A0_7501_4CDD_31E3.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

3. Choose the Partition Table Type

- TestDisk will automatically suggest the most likely partition table type (e.g., Intel/PC, EFI GPT).
- The default value is usually correct. Confirm the selection by pressing Enter.

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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - Micron MTFDKCD512TGE-1BK1AABLA

Please select the partition table type, press Enter when done.
>[Intel  ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax  ] Humax partition table
[Mac    ] Apple partition map (legacy)
[None   ] Non partitioned media
[Sun    ] Sun Solaris partition
[XBox   ] Xbox partition
[Return ] Return to disk selection_

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

4. Analyze Current Partition Structure

- From the main menu, choose [Analyse] and press Enter.
- This will display the current partition table and check for errors or missing partitions.

```
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - Micron MTFDKCD512TGE-1BK1AABLA

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection_

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

5. Perform a Quick Search

- After the analysis, you will be prompted to perform a [Quick Search].
- Select it and press Enter to scan the drive for lost partitions.

```

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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Current partition structure:
  Partition              Start      End      Size in sectors
1 P EFI System           2048      534527    532480 [EFI system partition]
No FAT, NTFS, ext2, JFS, Reiser, cramfs or XFS marker
2 P MS Reserved          534528    567295     32768 [Microsoft reserved partition]
2 P MS Reserved          534528    567295     32768 [Microsoft reserved partition]
No FAT, NTFS, ext2, JFS, Reiser, cramfs or XFS marker
3 P MS Data               567296    996118527 995551232 [Basic data partition]
3 P MS Data               567296    996118527 995551232 [Basic data partition]
4 P Windows Recovery Env  996118528 1000214527 4096000 [Basic data partition]

P=Primary D=Deleted
>[Quick Search] [ Backup ]
Try to locate partition

```

- Once a partition is found, you can press p to list its files. Deleted files and folders often appear in red.
- Press q to return to the search results.

6. Perform a Deeper Search

- If the Quick Search fails to find your lost partitions, select [Deeper Search]
- This process can take a long time, as it scans the entire drive, block by block, to find remnants of partition structures.

```

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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
>P EFI System    2048      534527    532480 [EFI System Partition] [SYSTEM_DRV]
D MS Data       992022529 996118528 4096000
D MS Data       996118528 1000214527 4096000 [WINRE_DRV]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
      P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
$FAT32, blocksizes=4096, 272 MB / 260 MiB

```

- Again, use p to preview files and confirm if a found partition is the one you are looking for.

7. Modify Partition Status

- After finding the correct partitions, use the Left/Right arrow keys to set their status.

```
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63

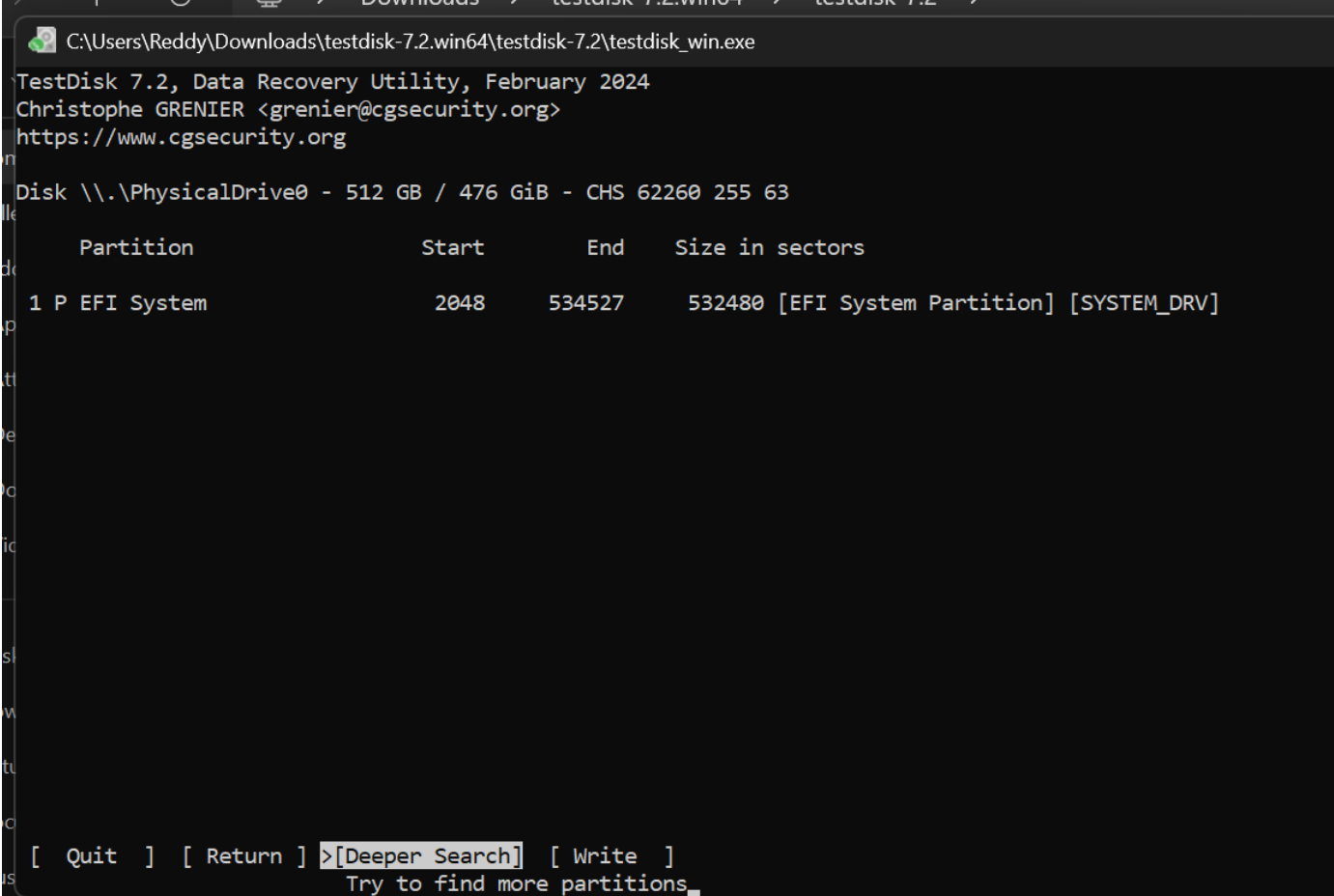
Partition              Start          End      Size in sectors
1 P EFI System          2048         534527    532480 [EFI System Partition] [SYSTEM_DRV]

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
                        Try to find more partitions_
```

- 8.
- Use **Left/Right arrow keys** to change status:
 - **P**: Primary
 - *****: Bootable
 - **L**: Logical
 - **D**: Deleted
 - ensure that the partitions you want to recover are marked as Primary or Logical (and not deleted).

9. Write the Partition Table

- Once you are confident the partition structure is correct, select [Write] from the menu.



The screenshot shows the TestDisk 7.2 interface. At the top, the file path is C:\Users\Reddy\Downloads\testdisk-7.2.win64\testdisk-7.2\testdisk_win.exe. Below this, the version and date are listed: TestDisk 7.2, Data Recovery Utility, February 2024. The author's name and email are Christophe GRENIER <grenier@cgsecurity.org>, and the website is https://www.cgsecurity.org. The disk information is: Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63. A table of partitions is shown with one entry: 1 P EFI System, starting at 2048 and ending at 534527, with a size of 532480 sectors. At the bottom, there are menu options: [Quit], [Return], >[Deeper Search], and [Write]. A prompt 'Try to find more partitions.' is also visible.

```
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63

Partition              Start      End      Size in sectors
1 P EFI System          2048      534527    532480 [EFI System Partition] [SYSTEM_DRV]

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
                        Try to find more partitions.
```

Confirm the operation by pressing y (for yes). This will write the new partition table to your disk.

```
C:\Users\Reddy\Downloads\testdisk-7.2.win64\testdisk-7.2\testdisk_win.exe
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Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - Micron MTFDKCD512TGE-1BK1AABLA
CHS 62260 255 63 - sector size=512

>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

10. Recover Files

- If you just need to recover a few files without fixing the partition table, you can do so from the file list (after pressing p).
- Navigate to the folder containing your desired files.
- Use the colon : key to select the files you want to recover.
- Press the uppercase C key to copy the selected file(s).

```
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Christophe GRENIER <grenier@cgsecurity.org>  
https://www.cgsecurity.org
```

```
Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
```

	Partition	Start	End	Size in sectors
>	1 P HPFS - NTFS	2 10 9	7791 180 5	125140992

```
[ Type ] >[ Boot ] [ List ] [Undelete] [Image Creation] [ Quit ]
```

- Navigate to a safe destination on a different storage device and press uppercase C again to paste.

11. Exit and Restart

- Once your task is complete, select [Quit] to exit the program.
- If you wrote a new partition table to the drive, it is recommended to restart your computer to allow the operating system to recognize the changes.

```
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https://www.cgsecurity.org
```

```
Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
```

Partition	Start	End	Size in sectors
1 P HPFS - NTFS	2 10 9	7791 180 5	125140992

```
Boot sector
```

```
exFAT OK
```

```
Backup boot record
```

```
exFAT OK
```

```
Sectors are not identical.
```

```
>[ Quit ] [Org. BS ] [Backup BS] [ Dump ]
```

References:

https://www.cgsecurity.org/wiki/TestDisk_Download

https://www.cgsecurity.org/testdisk_doc/